

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER V) END TERM EXAMINATION, JANUARY -2025****DATA VISUALIZATION USING R****[Time: 2 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Data Visualization Using R**

CO	Course Outcome (CO)
CO1	Understanding about the importance of packages in R and how they extend the functionality of the base language
CO2	Master the art of data visualization through comprehensive techniques, covering single and multiple variables, time series, and categorical data representation.
CO3	Excel in data visualization using ggplot2, mastering diverse plotting techniques and advanced functionalities for insightful analysis
CO4	Interpret classification and prediction methods to make accurate predictions, leading to better decision-making in various industries.

(Note- Faculty can add the more CO's as per their subject.)**SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)**

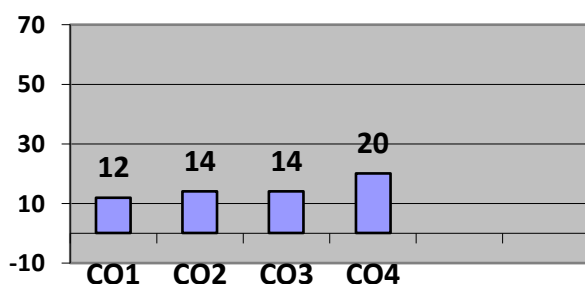
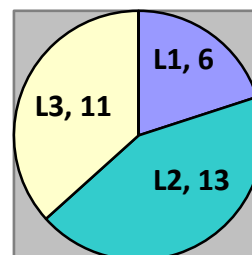
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	4	CO1	BL
(B)	4	CO2	BL
(C)	4	CO3	BL
(D)	4	CO1	BL
(E)	4	CO1	BL

SECTION-B**Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	10	CO2	BL
2 (B)	10	CO2	BL
3 (A)	10	CO3	BL
3 (B)	10	CO3	BL

SECTION-C**Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	10	CO4	BL
4 (B)	10	CO4	BL

Course Outcome Wise Marks Distribution**L1 L2 L3****Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER V) END TERM EXAMINATION, JANUARY - 2025

DATA VISUALIZATION USING R

[Time: 2 Hours]

[Maximum Marks: 60]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5×4=20)

- (a) Explain with a suitable example how a package is installed and loaded in R-Studio.
- (b) Differentiate between Discrete Variable and Continuous Variable.
- (c) What are the two types of functions in R? Explain with example.
- (d) Define Recursion. Is Recursion possible in R technology?
- (e) Write a code by hand on the answer-sheet (hardcopy) to convert the following vector “marks” in a 2 x 3 matrix row-wise.

marks←c(78, 77, 91, 90, 86, 66)

SECTION-B

(Practical / Descriptive Type Questions)

Q.2. Attempt all questions of the following

(2X10=20)

- (a) Using dataset Employee.csv construct a Box Whisker Plot on Salary. Identify the outliers, if any.
If the outliers are removed, then what is the shape of Salary Distribution? Justify your answer in hardcopy.

OR

- (b) Explain in brief with the help of suitable example about the R pipe operator.

Q.3. (a) Using dataset “iris” of R Base Package, construct Partitioned Density of the Sepal Length Plot across different species of iris flower. Interpret the outcome in hardcopy.

OR

- (b) Explain in brief about the significance of lattice package in Data Visualization using R.

SECTION-C

Q.4. Attempt all parts of the following:

(2×10=20)

Note: Use the dataset “diamonds” in ggplot2 package to attempt the following question Q.4. Use of package ggplot2 is mandatory but additional other packages are optional.

- 4(a)** Create a histogram of the values for price against each cut. Interpret the outcome in hardcopy while commenting on the shape of distribution.
- 4(b)** Create a scatter plot of Carat Size Versus Price using cut as color variable. Add a regression line to the scatter plot produced. Comment in the hardcopy whether there is an association between Carat Size and Prices on the basis of final outcome.